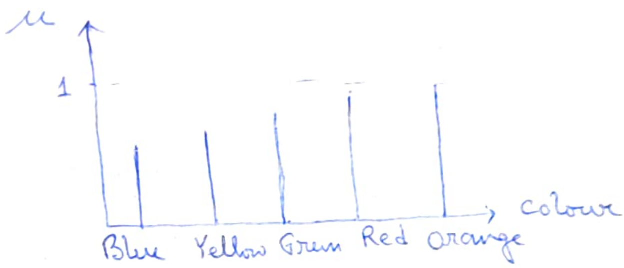
If the sum of preferences of each colour (row wise) is normalized to the total number of responses, a rank ordering can be determined as shown in last coloumn.

If the percentage preference plotted & of these colours to a normalized scale on the universe of Colours in an ascending order on the colour universe membership function for "best colour" would result.



<u>Angular Fuzzy Sets</u> -

Angular fuzzy sets only differs from standard fuzzy sets in their co-ordinate ~~positions~~ descriptions. Angular fuzzy sets are defined on universe of angles hence are repeating shapes every 2π cycles.

In most applications angular fuzzy sets are used in the quantitative descriptions of linguistic variables known as truth ~~table~~ values. when a certain proposition has a membership value of 1 it is said to be true and when the proposition membership value is 0 it is said to be false.

<u>Ex</u> - Measuring the pH level of polluted water

The pH levels are given linguistic ~~levels~~ labels.

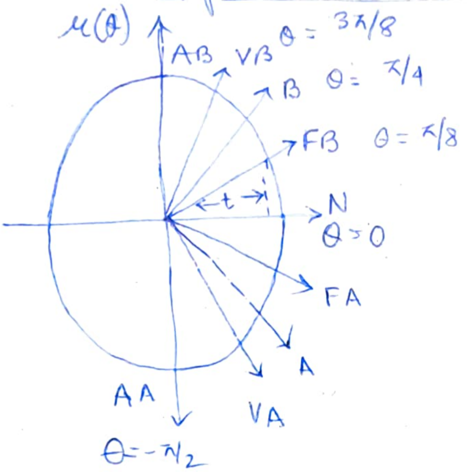
It is known that neutral solution has pH of 7.

Then linguistic term Neutral N $\theta = 0°$ rad

	Absolutely Basic	AB	$\theta = \pi/2$
	" Acidic	AA	$\theta = -\pi/2$
pH ~~label~~ level between 7 to 14	Very Basic	VB	$\theta = 3\pi/8$
	Fairly Basic	FB	$\theta = \pi/8$
	Basic	B	$\theta = \pi/4$

pH level is between 7 to 0 → FA, A, VA

The linguistic values vary with θ and their



membership values are on the μ(θ) axis.

Thus the membership value for the linguistic term can be obtained from

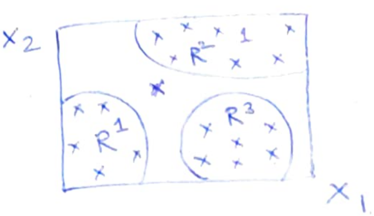
$$\mu_t(\theta) = t \cdot \tan(\theta)$$

where t is the horizontal projection of radial vector.

Membership function using Neural Net

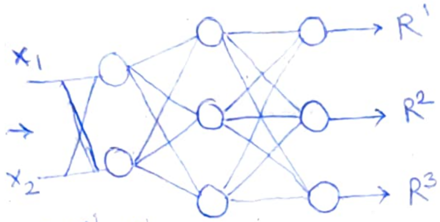
A number of input data set is selected among which a training set is selected and rest of the data is used for checking (testing).

Let, consider an example where the data points have been divided into three regions/classes R^1, R^2, R^3

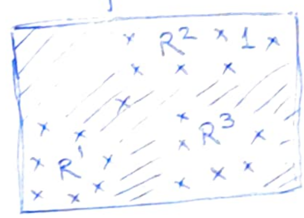


Let consider data point 1, which has co-ordinates $x_1 = 0.7$ and $x_2 = 0.8$. As this is in region R^2, we assign to it a complete membership 1 in class R^2 and 0 ~~to~~ membership to R^1 and R^3 classes.

A neural network is created that uses the data point 1 and corresponding membership values in different classes for training itself to simulate the relationship between coordinates locations and membership values.



Neural Network and its output



x_1:	0.7	0.8
x_2:	0.8	0.2

Input

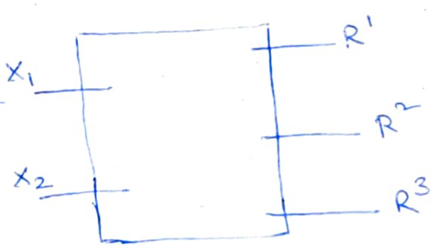
R^1	0	0
R^2	1	0
R^3	0	1

Output

In this method neural network is trained with all input data and corresponding classes. Then after training of neural network, it is used to determine ~~the~~ membership values of any input data.

A single data input

X_1	0.5
X_2	0.5



R^1	0.8
R^2	0.1
R^3	0.1

Membership values in each region for i/p data

Membership Function using GA

GA is also used to calculate membership function.

Some functional mapping for the system is given
↳ Some membership functions and shapes are assumed for the fuzzy variables as defined in the problem
↳ These membership functions are coded with binary bits/strings
↳ A fitness function is used to evaluate the fitness of each membership function

Ex -

Data for Single input Single output

x:	1	2	3	4	5
y:	1	4	9	16	25

Functional mapping for the system

x:	S	L
y:	S	VL

Each of the variables x and y makes use of two fuzzy classes.

x uses S (Small) and L (Large)

y uses ~~s~~ S (Small) and VL (very Large)

The functional mapping tells that a small x maps to a small y and a large x maps to very large y

Let, assume range of variables x [0, 5]
y [0, 25]

assume ~~the~~ each membership function has the shape of righ angle triangle.

Six bit string is used to define the base of each membership function

String Number	① String	② base(1) (bin)	③ base 2 (bin)	④ base3 (bin)	⑤ base 4 (bin)	⑥ base 1	⑦ base 2	⑧ base 3	⑨ base 4
	x(S) x(L) y(S) y(VL)								
1	000111 010100 010110 110011	7	20	22	51	0.56	1.59	8.73	20.24
2	010010 001100 101100 100110	18	12	44	38	1.43	0.95	17.46	15.08
3									
4									

The bit strings are of 24 bits length, where first 6 bits → x(S)
next 6 bits → y(S)
next 6 bits → x(L)
next 6 bits → y(VL)

x(S), x(L) → x(S, L)
y(S), y(VL) → y(S, VL)

The base value is calculated by using eqn $C_i = C_{min} + \frac{b}{2^L - 1}(C_{max_i} - C_{min_i})$

where b is the decimal value, corresponding to bit string (6 bits here)
L is the length of binary string
C_{min}, C_{max} are assumed according to the problem statements

Thus for string No. 1, small element of x, base 6

$$C_1 = 0 + \frac{7}{2^6 - 1}(5 - 0)$$ — Here $C_{min} = 0$ & $C_{max} = 5$ for membership functions S (Small) and L (Large) of x

$$= \frac{7}{63} \times 5 = 0.555$$

For string No. 1, very Large (VL) element of y, base 9

$$C_1 = 0 + \frac{51}{2^6 - 1}(25 - 0)$$ — Here $C_{min} = 0$ and $C_{max} = 5$ for S and VL of y

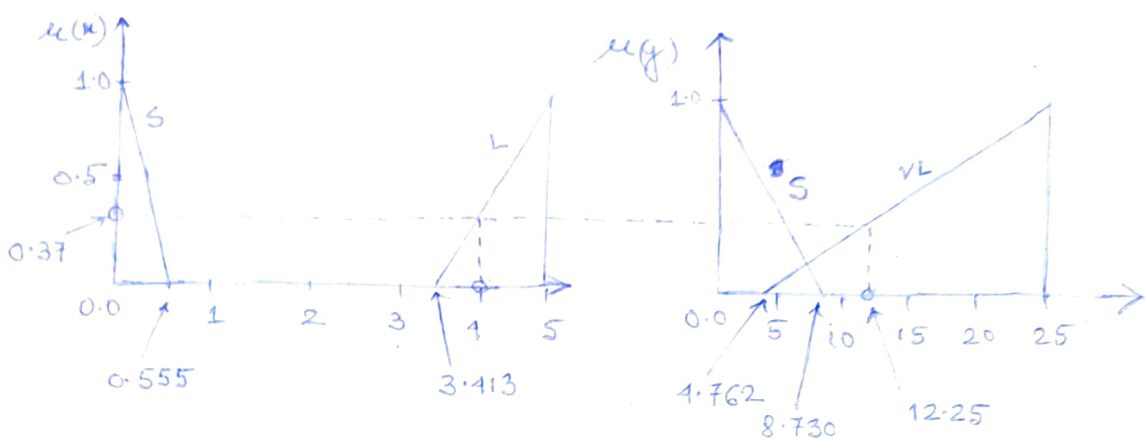
$$= 20.238$$

The strings are physically represented

Say for string No. 1:

values of x variables are 0.56 and $(5-1.59)=3.41$

values of y variables are 8.73 and $(25-20.24)=4.7$



Now the value of y can be calculated graphically from the membership function for string 1.

Say, for $x=4$, we see that the membership of x in the fuzzy class large is 0.37. and also ~~the~~ in the fuzzy class VL of fuzzy variable y is 12.25.

The corresponding actual value of y is 16,

$\therefore$ the squared error is $=(16-12.25)^2 = 14.06$.

Defuzzification

It has the result of reducing a fuzzy set to a crisp single ~~set~~ value or to a crisp set; of converting a fuzzy matrix to a crisp ~~mat~~ matrix; or making a fuzzy number to a crisp number.

Mathematically, the defuzzification of a fuzzy set is the process of "rounding it off" from its location in the unit hypercube to the nearest vertex. If fuzzy set is a collection of membership values, or a vector values on the unit interval, defuzzification reduces this vector to a single scalar quantity.

Lambda-cuts for Fuzzy Sets -

$\underset{\sim}{A} \rightarrow$ fuzzy set

Then Lambda-cut set, A_λ, where $0 \leq \lambda \leq 1$.

The set A_λ is crisp set is called Lambda-cut set of the fuzzy set $\underset{\sim}{A}$, where $A_\lambda = \{x \mid \mu_A(x) \geq \lambda\}$.

crisp set

~~Any~~ Any element $x \in A_\lambda$, belongs to $\underset{\sim}{A}$ with a grade of membership that is greater than or equal to value λ.

Ex - Let, us consider the discrete fuzzy set, using Zadeh's notation, defined on universe

$$X = \{a, b, c, d, e, f\}.$$

$$\underset{\sim}{A} = \left\{ \frac{1}{a} + \frac{0.9}{b} + \frac{0.6}{c} + \frac{0.3}{d} + \frac{0.01}{e} + \frac{0}{f} \right\}$$

Now this fuzzy set is reduced into several λ-cut sets, all of which are crisp.

$$\lambda = 1, 0.9, 0.6, 0.3, 0^+, \text{ and } 0$$

$A_1 = \{a\}$ $\qquad$ $A_{0.9} = \{a, b\}$

$A_{0.6} = \{a, b, c\}$ $\qquad$ $A_{0.3} = \{a, b, c, d\}$

$A_{0^+} = \{a, b, c, d, e\}$ $\qquad$ $A_0 = X$

Let us consider a fuzzy relation matrix

$$\underset{\sim}{R} = \begin{bmatrix} 1 & 0.8 & 0 & 0.1 & 0.2 \\ 0.8 & 1 & 0.4 & 0 & 0.9 \\ 0 & 0.4 & 1 & 0 & 0 \\ 0.1 & 0 & 0 & 1 & 0.5 \\ 0.2 & 0.9 & 0 & 0.5 & 1 \end{bmatrix}$$

$\underset{\sim}{R}$ fuzzy relation matrix is considered a fuzzy set.

Let us define $R_\lambda = \{(x, y) \mid \mu_{\underset{\sim}{R}}(x,y) \geq \lambda\}$ as a ~~&~~ λ-cut relation of the fuzzy relation, $\underset{\sim}{R}$.

Here $\underset{\sim}{R}$ is two dimensional array defined in universe X and Y ~~and~~ then any pair $(x,y) \in R_\lambda$ belongs to $\underset{\sim}{R}$ with a "strength" of relation greater than or equal to λ.

$\lambda = 1$ $\quad$ $R_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

$\lambda = 0.9$ $\quad$ $R_{0.9} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$

1. $(\underset{\sim}{A} \cup \underset{\sim}{B})_\lambda = A_\lambda \cup B_\lambda$
2. $(\underset{\sim}{A} \cap \underset{\sim}{B})_\lambda = A_\lambda \cap B_\lambda$
3. $(\bar{A})_\lambda \neq (\bar{A_\lambda})$ except for value $\lambda = 0.5$
4. For any $\lambda \leq \alpha$, where $0 \leq \alpha \leq 1$, it is true that $A_\alpha \subseteq A_\lambda$ where $A_0 = X$

1. $(\underset{\sim}{R} \cup \underset{\sim}{S}) = R_\lambda \cup S_\lambda$
2. $(\underset{\sim}{R} \cap \underset{\sim}{S})_\lambda = R_\lambda \cap S_\lambda$
3. $(\bar{\underset{\sim}{R}})_\lambda \neq \bar{R}_\lambda$
4. For any $\lambda \leq \alpha$, $0 \leq \alpha \leq 1$ then $R_\alpha \subseteq R_\lambda$

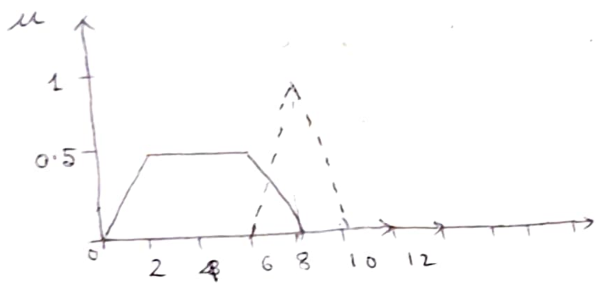
Deffuzzification methods

~~Deff~~ Defuzzification is the conversion of a fuzzy quantity ~~as opposed~~ to a precise quantity. The ~~quantity~~ output of a fuzzy process can be the logical union of two or more fuzzy membership functions defined on the universe of ~~discrete sure~~ discourse of the output variable.

For example, suppose a fuzzy output is comprised of two parts C_1 : Trapezoidal shape
C_2 : Triangular shape

$\left.\begin{matrix} \\ \\ \end{matrix}\right\}$ union of these two function $\underset{\sim}{C} = \underset{\sim}{C_1} \cup \underset{\sim}{C_2}$

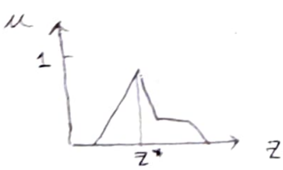
→ This includes max- operator



The popular defuzzification output-functions are -

1. <u>Max-membership principle</u>

Also known as the height method, this scheme is limited to ~~Picked~~ peaked output functions.

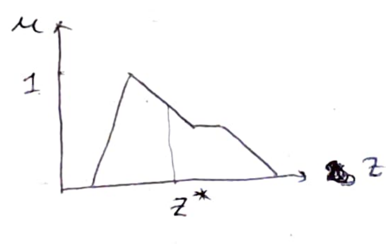


$$\mu_{\underset{\sim}{C}}(z^*) \geq \mu_{\underset{\sim}{C}}(z) \quad \text{for all } z \in Z$$

2. <u>Centroid method</u>

It is also known as center of area, center of gravity is most prevalent defuzzification method.

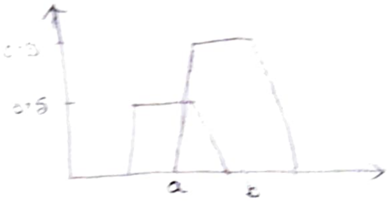
$$z^* = \frac{\int \mu_{\underset{\sim}{C}}(z) \cdot z \, dz}{\int \mu_c(z) \, dz}$$



3. Weighted average method

This method is only valid for symmetrical output membership function.

$$z^* = \frac{\sum \mu_{\underset{\sim}{c}}(\bar{z}) \cdot \bar{z}}{\sum \mu_{\underset{\sim}{c}}(\bar{z})}$$



This method is formed by weighting each membership function in the output by its respective maximum membership value.

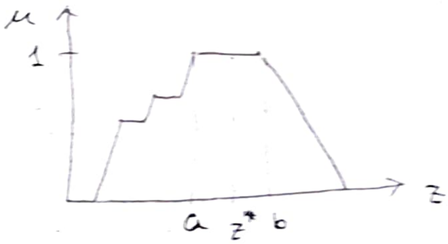
For the above example $z^* = \frac{(0.5)a + (0.9)b}{0.5 + 0.9}$

Here 'a' and 'b' are means of their respective shapes

4. Mean - max membership

This method is closely related to max-membership ~~ex~~ except that the location of the maximum membership can be non-unique.

$$z^* = \frac{a+b}{2}$$



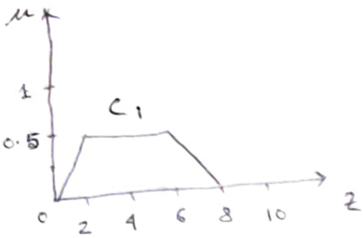
5. Center of sums

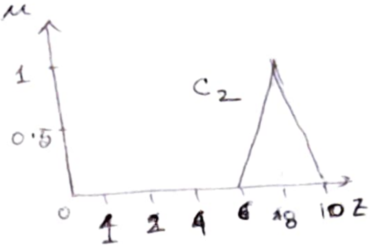
This is a faster ~~fuzzification~~ ~~method~~ defuzzification method which involves the algebraic sum of individual output fuzzy sets, say $\underset{\sim}{C}_1$ and $\underset{\sim}{C}_2$ instead of union.

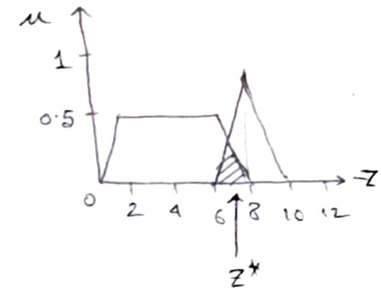
$$z^* = \frac{\int_z z\left(\sum_{k=1}^{n} \mu_{\underset{\sim}{C}_k}(z)\, dz\right)}{\int_z \sum_{k=1}^{n} \mu_{\underset{\sim}{C}_k}(z)\, dz}$$

One drawback of doing algebraic sum over union is that intersecting areas are added twice.

center of sum method differs from weighted average method in terms of weights which are the areas of respective membership functions instead of individual membership values.



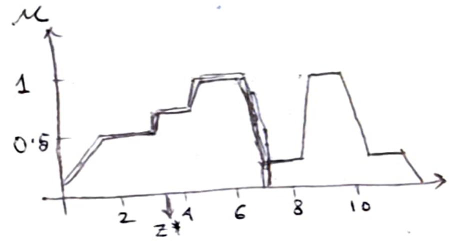




6. <u>Center of largest area</u>

If the output fuzzy set has atleast two convex subregions, then the center of gravity (i.e., z^* is calculated using the centroid method) of the convex fuzzy subregion with the largest area is used to obtain the defuzzification value z^* of the output.

$$z^* = \frac{\int \mu_{\underset{\sim}{C}m}(z)\, z\, dz}{\int \mu_{\underset{\sim}{C}m}(z)\, dz}$$



where C_m is the convex subregion that has the largest area making up $\underset{\sim}{C}_K$. This condition applies when the overall output C_K is non-convex.

7. <u>First (or Last) maxima</u>

This method uses the overall output or union of all individual output fuzzy sets $\underset{\sim}{C}_K$ to determine the smallest value of the domain with maximized membership degree in C_K.

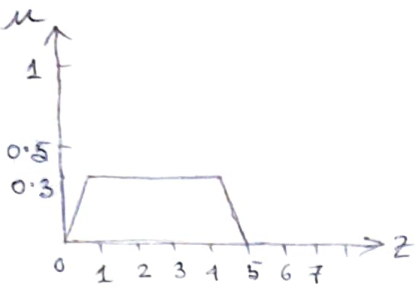
First, the largest height of the union is determined

$$hgt(\underset{\sim}{C}_K) = \sup_{z \in Z} \mu_{C_K}(z)$$

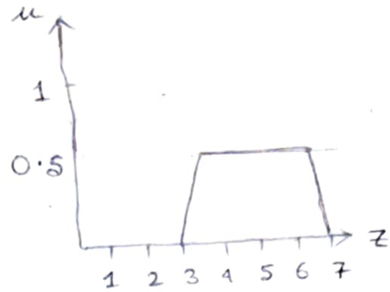
then first maxima

$$z^* = \inf_{z \in Z} \{ z \in Z \mid \mu_{C_k}(z) = hgt(C_k) \}$$

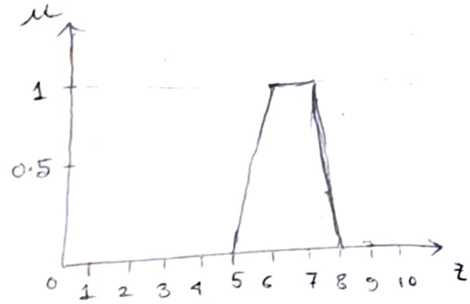
Ex - Let us consider $\tilde{A}_1, \tilde{A}_2, \tilde{A}_3$ are three fuzzy sets



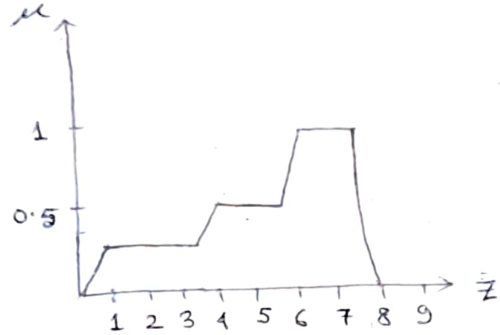
$\tilde{A}_1$



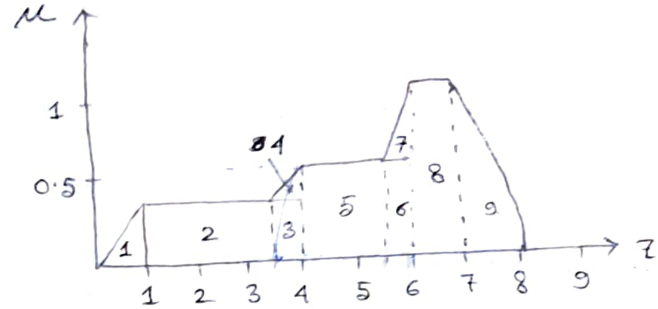
$\tilde{A}_2$



$\tilde{A}_3$



Aggregate fuzzy set $\tilde{A}_1, \tilde{A}_2, \tilde{A}_3$



Area No.	Area	$\bar{z}$	$A\bar{z}$
1.	$\frac{1}{2} \times 0.3 \times 1 = 0.15$ $(\frac{1}{2} a \times b)$	0.67 $(\frac{2}{3}a)$	0.1005
2.	$2.6 \times 0.3 = 0.78$	2.3	1.794
3.	$0.3 \times 0.4 = 0.12$	3.8	0.456
4.	$\frac{1}{2} \times 0.4 \times 0.2 = 0.04$	3.8667	0.1546
5.	$1.5 \times 0.5 = 0.75$	4.75	3.5625
6.	$0.5 \times 0.5 = \cancel{0.125}$ 0.25	~~5.8333~~ 5.75	~~0.729~~ 1.4375
7.	$\frac{1}{2} \times 0.5 \times 0.5 = 0.125$	5.833	0.729
8.	$1 \times 1 = 1$	6.5	6.5
9.	$\frac{1}{2} \times 1 \times 1 = 0.5$	7.33	3.665

* $\bar{z}$ for right angle triangle $= \frac{ab}{3}$.

1. $\bar{z} = \frac{2 \times 1}{3} = 0.67$

4. $\bar{z} = \frac{(4 - 3.6) \times 2}{3} + 3.6 = \frac{0.4 \times 2}{3} + 3.6 = 0.266 + 3.6 = 3.866$

Centroid method

$$z^* = \left[\int_0^1 (0.3z)\,z\,dz + \int_1^{3.6} (0.3z)\,dz + \int_{3.6}^{4} \left(\frac{z-3}{2}\right) z\,dz + \int_4^{5.5} (0.5z)\,dz + \int_{5.5}^{6} (z-5)\,z\,dz + \int_6^7 z\,dz + \int_7^8 (8-z)\,z\,dz\right] \div \left[\int_0^1 (0.3z)\,dz + \int_1^{3.6} (0.3)\,dz + \int_{3.6}^{4} \left(\frac{z-3}{2}\right) dz + \int_4^{5.5} (0.5z)\,dz + \int_{5.5}^{6} (z-5)\,dz + \int_6^7 dz + \int_7^8 (8-z)\,dz\right]$$

$$z^* = \frac{\sum A\bar{z}}{\sum A} = \frac{18.353}{3.715} = 4.9$$

Weighted average method

$$z^* = \frac{(0.3\times 2.5) + (0.5\times 5) + (1\times 6.5)}{0.3 + 0.5 + 1} = 5.41$$

Center of sums

$$z^* = \frac{\frac{1}{2}\times 0.3\times(3+5)\times 2.5 + \frac{1}{2}\times 0.5\times(4+2)\times 5 + \frac{1}{2}\times 1\times(3+1)\times 6.5}{\frac{1}{2}\times 0.3\times(3+5) + \frac{1}{2}\times 0.5\times(4+2) + \frac{1}{2}\times 1\times(3\times 1)}$$

$= 5.042$

$\frac{1}{2}\times h\times(a+b)$

$\frac{1}{2}\times 0.3\times(3+5)$ — area covered by A_1 and 2.5 is centroid

$\frac{1}{2}\times 0.5\times(4+2)$ — " " " A_2 ~ 5 " "

$\frac{1}{2}\times 1\times(3+1)$ — " " " A_3 ~ 6.5 " "

Mean-max membership

Since the aggregated fuzzy set is a continuous set, z^* is calculated the mean of maxima as $z^* = 6.5$

$M = \{z \in [6,7] \mid \mu(z) = 1\}$ and the height of the aggregated fuzzy set is 1.

Center of Largest area

It provides same result $(z^* = 4.9)$ as centroid method, because complete area of output fuzzy set is convex.